

A 3D Manifold Nodal Discontinuous Galerkin Method for Advection on the Octahedral Sphere

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摘要: 第一段移到 Introduction.
二、三、四段浓缩合併, 只要說明
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Abstract

High-order numerical simulations on spherical geometries often suffer from geometric singularities associated with conventional latitude-longitude parameterizations. In particular, equal-area mappings may simplify the metric Jacobian but still generate singular or ill-conditioned contravariant metric terms near polar regions, leading to unstable velocity projections and degraded numerical robustness.

To address these difficulties, this work develops a three-dimensional Manifold Discontinuous Galerkin (Manifold DG) method formulated directly on curved surfaces embedded in $\mathbb{R}^3$. A subdivided Octahedral mesh is employed to avoid pole singularities and multi-patch coordinate splicing. All geometric quantities, including covariant basis vectors, contravariant basis vectors, and surface Jacobians, are constructed intrinsically from the embedded surface geometry through differential geometric formulations.

The scalar advection equation on the spherical manifold is discretized using a nodal DG formulation with upwind numerical fluxes. A continuous energy analysis is presented under suitable regularity assumptions on the velocity field, while the semi-discrete formulation is analyzed through an approximate SBP framework on curved manifold elements. Particular attention is devoted to the influence of geometric aliasing errors and discrete metric inconsistencies on the discrete geometric conservation law (DGCL).

Numerical experiments involving solid-body rotation demonstrate stable fourth- to fifth-order convergence while effectively avoiding polar singularities. The numerical results further suggest that exact discrete conservation is limited by geometric truncation errors, non-polynomial metric terms, and imperfect discrete metric identities on curved manifold meshes. These observations provide insight into the construction of future entropy-stable and strictly conservative DG formulations on curved geometries.

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Chapter 1

Introduction

High-order numerical methods on spherical geometries play an important role in computational fluid dynamics, atmospheric modeling, climate simulations, and global wave propagation problems. In these applications, the accurate numerical treatment of transport phenomena on curved manifolds is essential for maintaining long-time stability, minimizing numerical dissipation, and preserving physically meaningful solution structures.

Among various high-order discretization techniques, the Discontinuous Galerkin (DG) method has attracted considerable attention due to its spectral-like accuracy, compact stencil structure, geometric flexibility, and strong parallel scalability. These advantages make DG methods particularly suitable for solving hyperbolic conservation laws on unstructured meshes and complex geometries.

Despite these benefits, constructing stable and accurate DG formulations on spherical geometries remains highly challenging. Traditional latitude–longitude coordinate systems inherently contain pole singularities, which introduce severe numerical instabilities near the North and South Poles. In addition, parameterized spherical mappings often require complicated topological splicing procedures across periodic boundaries or multi-patch interfaces, substantially increasing the complexity of geometric metric evaluations and numerical flux constructions.

In previous studies, we attempted to solve spherical advection equations by combining simplex quadrature, Octahedral meshes, and equal-area mappings from spherical coordinates onto planar computational domains. Although the equal-area mapping simplifies the Jacobian into a constant form, the associated contravariant metric terms become geometrically singular near the polar regions. Consequently, the projected contravariant velocities become ill-conditioned, leading to unstable numerical flux evaluations and degraded robustness of the DG discretization.

Rather than attempting to repair the singular parameterization framework, this work abandons the traditional two-dimensional surface parameterization approach and reformulates the DG discretization directly on an embedded manifold surface in $\mathbb{R}^3$. To achieve this goal, this study develops a three-dimensional Manifold Discontinuous Galerkin (Manifold DG) formulation operating entirely in Euclidean space. A subdivided Octahedral mesh is employed to construct the spherical geometry, thereby avoiding coordinate singularities and complicated periodic patch

connections.

Within this framework, all geometric quantities are computed intrinsically from the embedded surface representation. Covariant basis vectors, surface Jacobians, and contravariant metric terms are derived directly from the three-dimensional geometry, enabling physical velocity fields to be consistently projected onto local tangent spaces.

The governing advection equation is discretized using a nodal DG formulation with upwind numerical fluxes. A continuous energy analysis is first performed to establish conditional well-posedness of the continuous problem under suitable regularity assumptions. Subsequently, a discrete energy analysis is developed to investigate the stability properties of the semi-discrete formulation and identify the geometric conditions required for discrete conservation on curved manifold meshes.

Chapter.

The remainder of this report is organized as follows. ~~Section~~ 2.1 introduces the non-dimensionalization framework. ~~Section~~ 2.2 presents the continuous model problem and its energy estimate. Chapter 3 describes the numerical discretization procedures, including geometric mapping, matrix operators, spatial discretization, discrete energy stability analysis, and time integration methods. Chapter 4 presents numerical experiments and convergence studies for solid-body rotation test cases. Finally, Chapter 5 summarizes the main findings and discusses the relationship between geometric aliasing, discrete metric identities, and DGCL errors on curved manifold meshes.

簡述成果：是否達到預期。

Chapter 2

Mathematical Formulation

2.1 Non-dimensionalization

To enhance numerical stability and avoid floating-point truncation errors associated with large physical scales (e.g., the Earth's radius), the governing equation is non-dimensionalized prior to numerical discretization.

First, we define the characteristic scales of the system:

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- **Characteristic length:** $L_c = R$ (the physical radius of the sphere).
- **Characteristic velocity:** $U_c = R\omega$ (the maximum linear rotational velocity at the equator, where ω is the angular velocity).
- **Characteristic time:** $T_c = L_c/U_c = 1/\omega$ (the time required for a fluid parcel to travel one characteristic length).

Next, we introduce the dimensionless variables (denoted by an asterisk *):

$$\mathbf{x}^* = \frac{\mathbf{x}}{R}, \quad \mathbf{u}^* = \frac{\mathbf{u}}{U_c}, \quad t^* = \frac{t}{T_c}. \quad (2.1)$$

Substituting these variables into the original advection equation, and noting that the spherical divergence operator satisfies $\nabla_{\mathbb{S}} = \frac{1}{R}\nabla_{\mathbb{S}^*}$, we obtain:

$$\frac{1}{T_c} \frac{\partial q}{\partial t^*} + \frac{1}{R} \nabla_{\mathbb{S}^*} \cdot (U_c \mathbf{u}^* q) = 0. \quad (2.2)$$

Multiplying the entire equation by T_c and applying $T_c U_c / R = 1$ yields the non-dimensionalized conservation law:

$$\frac{\partial q}{\partial t^*} + \nabla_{\mathbb{S}^*} \cdot (\mathbf{u}^* q) = 0. \quad (2.3)$$

Through this transformation, all subsequent geometric metrics, numerical fluxes, and basis functions are uniformly constructed on the standard unit sphere ($\mathbb{S}^2$ with $R = 1$). This ensures that geometric terms (such as the Jacobian) remain at an $\mathcal{O}(1)$ magnitude, significantly improving

the robustness of the numerical framework. For simplicity of notation, the asterisk superscript (*) will be omitted in the subsequent mathematical formulations unless specified otherwise.

2.2 Well-posedness

We consider the scalar transport equation on the complete sphere $\mathbb{S}$:

$$\frac{\partial q}{\partial t} + \nabla_{\mathbb{S}} \cdot (\mathbf{u}q) = 0, \quad \mathbf{x} \in \mathbb{S}, \quad t \geq 0. \quad (2.4)$$

together with the initial condition

$$q(\mathbf{x}, 0) = q_0(\mathbf{x}). \quad (2.5)$$

Since the problem is posed on the complete sphere, there is no physical boundary and thus no boundary conditions are required. Throughout this section, we assume the tangential velocity field satisfies

$$\mathbf{u} \in W^{1,\infty}(\mathbb{S} \times [0, T]), \quad \nabla_{\mathbb{S}} \cdot \mathbf{u} \in L^{\infty}(\mathbb{S} \times [0, T]). \quad (2.6)$$

The initial-value problem is considered well-posed in the sense of Hadamard if:

1. A solution exists,
2. The solution is unique,
3. The solution continuously depends on the initial data.

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1. Existence

Under the assumption of a sufficiently smooth tangential velocity field $\mathbf{u}$, the existence of a classical solution can be constructed through the characteristic flow. Consider the trajectory $X(t)$ satisfying

$$\frac{dX(t)}{dt} = \mathbf{u}(X(t), t), \quad X(0) = X_0 \in \mathbb{S}. \quad (2.7)$$

Since $\mathbf{u}$ is tangent to the sphere, the characteristic trajectories remain on $\mathbb{S}$ for all time. Along these characteristic curves, the transport equation reduces to an ordinary differential equation, allowing the solution to be propagated continuously from the initial data q_0 . Therefore, a classical solution exists on the sphere under the stated regularity assumptions.

2. Stability (Energy Estimate)

Define the continuous L^2 energy

$$E(t) = \|q(\cdot, t)\|_{L^2(\mathbb{S})}^2 = \int_{\mathbb{S}} q^2 dS. \quad (2.8)$$

Differentiating with respect to time yields

$$\frac{dE}{dt} = 2 \int_{\mathbb{S}} q \partial_t q dS. \quad (2.9)$$

Substituting the governing equation gives

$$\frac{dE}{dt} = -2 \int_{\mathbb{S}} q \nabla_{\mathbb{S}} \cdot (\mathbf{u}q) dS. \quad (2.10)$$

Using the product rule,

$$\nabla_{\mathbb{S}} \cdot (q^2 \mathbf{u}) = 2q \nabla_{\mathbb{S}} \cdot (\mathbf{u}q) - q^2 \nabla_{\mathbb{S}} \cdot \mathbf{u}, \quad (2.11)$$

we obtain

$$\frac{dE}{dt} = - \int_{\mathbb{S}} \nabla_{\mathbb{S}} \cdot (q^2 \mathbf{u}) dS - \int_{\mathbb{S}} q^2 \nabla_{\mathbb{S}} \cdot \mathbf{u} dS. \quad (2.12)$$

Since the sphere has no boundary, the surface divergence theorem gives

$$\int_{\mathbb{S}} \nabla_{\mathbb{S}} \cdot (q^2 \mathbf{u}) dS = 0. \quad (2.13)$$

Therefore,

$$\frac{dE}{dt} = - \int_{\mathbb{S}} q^2 \nabla_{\mathbb{S}} \cdot \mathbf{u} dS. \quad (2.14)$$

~~Define~~ **ASSUME THAT $\nabla_{\mathbb{S}} \cdot \mathbf{u}$ IS BOUNDED:**

$$\alpha = \|\nabla_{\mathbb{S}} \cdot \mathbf{u}\|_{L^\infty(\mathbb{S} \times [0, T])}. \quad (2.15)$$

Then

$$\frac{dE}{dt} \leq \alpha E(t). \quad (2.16)$$

Applying Grönwall's inequality yields

$$E(t) \leq e^{\alpha t} E(0). \quad (2.17)$$

This fundamental energy bound will be used to establish both uniqueness and continuous dependence on the initial data.

3. Uniqueness

Suppose q_1 and q_2 are two solutions satisfying the same initial condition. Define

$$w = q_1 - q_2. \quad (2.18)$$

Then w satisfies

$$\partial_t w + \nabla_{\mathbb{S}} \cdot (\mathbf{u}w) = 0, \quad w(\mathbf{x}, 0) = 0. \quad (2.19)$$

Applying the energy estimate derived above yields

$$\|w(\cdot, t)\|_{L^2(\mathbb{S})}^2 \leq e^{\alpha t} \|w(\cdot, 0)\|_{L^2(\mathbb{S})}^2 = 0. \quad (2.20)$$

Hence,

implying that $q_1 = q_2$ at all time. This ~~improves~~ $w(\mathbf{x}, t) \equiv 0$, (2.21)
~~which~~ proves uniqueness.

4. Continuous Dependence (Perturbation Analysis)

To show that the solution is stable and depends continuously on the initial data, we consider an unperturbed problem and a perturbed problem. Let $q(\mathbf{x}, t)$ be the solution to the **unperturbed problem**:

$$\frac{\partial q}{\partial t} + \nabla_{\mathbb{S}} \cdot (\mathbf{u}q) = 0, \quad q(\mathbf{x}, 0) = q_0(\mathbf{x}). \quad (2.22)$$

Let $\tilde{q}(\mathbf{x}, t)$ be the solution to the **perturbed problem** with a small initial perturbation $\epsilon(\mathbf{x})$:

$$\frac{\partial \tilde{q}}{\partial t} + \nabla_{\mathbb{S}} \cdot (\mathbf{u}\tilde{q}) = 0, \quad \tilde{q}(\mathbf{x}, 0) = q_0(\mathbf{x}) + \epsilon(\mathbf{x}). \quad (2.23)$$

Let $w(\mathbf{x}, t) = \tilde{q}(\mathbf{x}, t) - q(\mathbf{x}, t)$ be the difference between the perturbed and unperturbed solutions. Due to the linearity of the transport equation, w satisfies:

$$\frac{\partial w}{\partial t} + \nabla_{\mathbb{S}} \cdot (\mathbf{u}w) = 0, \quad w(\mathbf{x}, 0) = \epsilon(\mathbf{x}). \quad (2.24)$$

Applying the energy estimate derived in Section 2 to $w(\mathbf{x}, t)$, we obtain:

$$\|w(\cdot, t)\|_{L^2(\mathbb{S})}^2 \leq e^{\alpha t} \|w(\cdot, 0)\|_{L^2(\mathbb{S})}^2, \quad (2.25)$$

which is equivalent to:

$$\int_{\mathbb{S}} (\tilde{q}(\mathbf{x}, t) - q(\mathbf{x}, t))^2 dS \leq e^{\alpha t} \int_{\mathbb{S}} \epsilon^2(\mathbf{x}) dS. \quad (2.26)$$

This implies that the difference between the perturbed solution $\tilde{q}$ and the unperturbed solution q

is bounded by the initial perturbed data $\epsilon(\mathbf{x})$ amplified by a factor of $e^{\alpha t}$. For any finite time T , if the initial perturbation $\|\epsilon\|_{L^2(\mathbb{S})} \ll 1$, the deviation of the solution remains small. Therefore, the system is stable and continuously depends on the initial data.

Chapter 3

Numerical Methods

3.1 Mapping to the Octahedral Sphere

這部內容的參考
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To map the computational domain onto the sphere, a direct Euclidean geometric projection method is employed to maintain numerical symmetry while avoiding pole singularities. Consider a triangular element on the Octahedral mesh with physical vertices

$$\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3.$$

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說明？

(3.1)

Using barycentric coordinates on the reference triangle (ξ, η) :

$$L_1 = \frac{-\xi - \eta}{2}, \quad L_2 = \frac{\xi + 1}{2}, \quad L_3 = \frac{\eta + 1}{2}, \quad (3.2)$$

the interpolated flat coordinate is

$$\mathbf{x}_{\text{flat}} = L_1 \mathbf{v}_1 + L_2 \mathbf{v}_2 + L_3 \mathbf{v}_3. \quad (3.3)$$

Based on the non-dimensionalization framework established in Chapter 2, the point is projected onto the standard unit sphere ($R = 1$):

$$\mathbf{x}_s = \frac{\mathbf{x}_{\text{flat}}}{\|\mathbf{x}_{\text{flat}}\|}. \quad (3.4)$$

This ensures that the covariant basis vectors and surface Jacobian are pure geometric ratios, entirely decoupling the influence of physical spatial scales from the numerical metrics.

Since this mapping contains a nonlinear normalization operator, the resulting geometry is non-polynomial and rational in (ξ, η) . Consequently, metric quantities are not exactly representable in the finite polynomial approximation space.

The covariant basis vectors are defined as

$$\mathbf{a}_1 = \frac{\partial \mathbf{x}_s}{\partial \xi}, \quad \mathbf{a}_2 = \frac{\partial \mathbf{x}_s}{\partial \eta}. \quad (3.5)$$

In the present implementation, these derivatives are evaluated numerically through the DG differentiation operators applied to nodal geometric coordinates. Therefore, the resulting metric terms are projected polynomial approximations rather than exact analytic derivatives. The unnormalized surface normal vector is

$$\mathbf{N} = \mathbf{a}_1 \times \mathbf{a}_2, \quad (3.6)$$

and the surface Jacobian is

$$J = \|\mathbf{N}\|. \quad (3.7)$$

The unit normal vector becomes

$$\mathbf{n}_{\text{surf}} = \frac{\mathbf{N}}{J}. \quad (3.8)$$

The contravariant basis vectors are then constructed as

$$\mathbf{a}^1 = \frac{\mathbf{a}_2 \times \mathbf{n}_{\text{surf}}}{J}, \quad \mathbf{a}^2 = \frac{\mathbf{n}_{\text{surf}} \times \mathbf{a}_1}{J}. \quad (3.9)$$

Given a physical velocity field

$$\mathbf{u} = (u_x, u_y, u_z) \in \mathbb{R}^3, \quad (3.10)$$

its contravariant velocity components are defined through tangent-space projection:

$$\tilde{u} = \mathbf{a}^1 \cdot \mathbf{u}, \quad \tilde{v} = \mathbf{a}^2 \cdot \mathbf{u}. \quad (3.11)$$

This formulation corresponds to a geometric projection of the physical velocity onto the local contravariant tangent basis induced by the embedded manifold geometry.

3.2 Reference Element and Matrix Operators

Following the standard nodal DG framework of Hesthaven and Warburton [3], we first define the discrete spaces and matrix operators on the reference element.

1. Nodal Basis and Mass Matrix

On the reference triangle, we employ nodal Lagrange basis functions $\phi_i(\xi, \eta)$. Quadrature rules from Chen and Shu [2] are used to construct discrete operators. Let $\mathbf{V}$ denote the Vandermonde matrix

$$\mathbf{V}_{mi} = \phi_i(\xi_m, \eta_m), \quad (3.12)$$

and let $\mathbf{W}$ denote the diagonal quadrature matrix.

The mass matrix is

$$\mathbf{M} = |T| \mathbf{V}^T \mathbf{W} \mathbf{V}. \quad (3.13)$$

2. Differentiation Matrices

The differentiation matrices are defined as

$$\mathbf{D}_\xi = |T| \mathbf{V}_\xi \mathbf{M}^{-1} \mathbf{V}^T \mathbf{W}, \quad \mathbf{D}_\eta = |T| \mathbf{V}_\eta \mathbf{M}^{-1} \mathbf{V}^T \mathbf{W}. \quad (3.14)$$

3. Face Extraction and Boundary Integration

The extraction matrix $\mathbf{E}$ maps volume nodes onto face nodes:

$$\mathbf{E} = [I_{3(n+1) \times 3(n+1)} \mid 0]. \quad (3.15)$$

The boundary quadrature matrix $\mathbf{W}_b$ contains the edge quadrature weights.

3.3 Spatial Discretization

Consider the conservation law

$$\frac{\partial q}{\partial t} + \nabla_{\mathbb{S}} \cdot (\mathbf{u}q) = 0. \quad (3.16)$$

1. Volume Term

Using the projected contravariant velocities $(\tilde{u}, \tilde{v})$, the volume operator is written as

$$\mathcal{L}_{\text{vol}}(q) = \frac{1}{J} [\mathbf{D}_\xi (J\tilde{u} \odot q) + \mathbf{D}_\eta (J\tilde{v} \odot q)]. \quad (3.17)$$

Because the metric Jacobian and contravariant velocity components are non-polynomial functions on curved elements, the products

$$J\tilde{u}q, \quad J\tilde{v}q \quad (3.18)$$

generally exceed the exact representability of the degree- k nodal polynomial space. Consequently, geometric aliasing errors are introduced into the discrete differentiation operators.

Proposition 2.1 (Geometric consistency limitation)

On curved manifold DG discretizations constructed from non-polynomial geometric mappings, the discrete SBP identity is generally violated by geometric approximation errors induced by projected metric terms and nonlinear geometric products. In sufficiently smooth regions, these geometric inconsistencies are expected to decay asymptotically with mesh refinement at rates consistent with the underlying polynomial approximation accuracy, although the precise convergence behavior additionally depends on quadrature accuracy, metric projections, and geometric aliasing effects.

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2. Surface Flux

The effective normal velocity is defined as

$$u_n = J^{-1} (n_\xi \tilde{u} + n_\eta \tilde{v}), \quad (3.19)$$

where (n_ξ, n_η) denotes the outward unit normal in the reference coordinate system.

Let

$$f = u_n q^-. \quad (3.20)$$

The upwind numerical flux is defined by

$$f^*(q^-, q^+) = \frac{u_n q^- + u_n q^+}{2} + \frac{|u_n|}{2}(q^- - q^+). \quad (3.21)$$

The penalty term is

$$p = f - f^*. \quad (3.22)$$

The surface correction operator becomes

$$\mathcal{L}_{\text{surf}}(q) = \mathbf{V}\mathbf{M}^{-1}\mathbf{V}^T\mathbf{E}^T\mathbf{W}_b p. \quad (3.23)$$

Combining the volume and surface terms yields the semi-discrete system:

$$\frac{\partial q}{\partial t} = -\frac{1}{J} [\mathbf{D}_\xi(J\tilde{u} \odot q) + \mathbf{D}_\eta(J\tilde{v} \odot q)] + \mathbf{V}\mathbf{M}^{-1}\mathbf{V}^T\mathbf{E}^T\mathbf{W}_b p. \quad (3.24)$$

3.4 A Conditional Discrete Energy Estimate

有什么条件？
需要什么条件？

In this section, we analyze the semi-discrete energy behavior of the proposed Manifold DG scheme. We adopt a conditional approach: first establishing an energy estimate under ideal discrete geometric compatibility, and subsequently identifying the defect terms that arise in the actual implementation on curved manifolds.

1. Semi-discrete Formulation and Discrete Energy Norm

To establish a baseline for stability, we first define the discrete energy norm and assume that the discrete operators perfectly satisfy ideal conditions on the manifold. Let $\mathbf{q}$ denote the vector of nodal unknowns on an element T . For a given surface element T , the discrete energy norm is defined as:

$$\|q\|_T^2 = \mathbf{q}^T \mathbf{W} \mathbf{J} \mathbf{q}, \quad (3.25)$$

where $\mathbf{W}$ is the diagonal matrix of quadrature weights, and $\mathbf{J}$ is the diagonal matrix of the metric Jacobian. This discrete norm is equivalent to the continuous integral $\int_T q^2 d\Omega$.

The semi-discrete formulation can be abstracted as:

$$\mathbf{J} \frac{d\mathbf{q}}{dt} + \mathcal{L}_{\text{vol}}(\mathbf{q}) - \mathbf{V}\mathbf{M}^{-1}\mathbf{V}^T\mathbf{E}^T\mathbf{W}_b(f - f^*) = \mathbf{0}. \quad (3.26)$$

2. Core Ideal Assumptions

The ideal discrete energy analysis relies on three core assumptions:

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- **Assumption 1 (Exact SBP Property):** The differentiation matrices satisfy the Summation-by-Parts (SBP) property.
- **Assumption 2 (Discrete Incompressibility):** The numerical operators strictly preserve the incompressibility condition ($\nabla \cdot V = 0$) at the discrete level.
- **Assumption 3 (Anti-symmetric Normal Velocity):** The interface normal velocity is anti-symmetric between adjacent elements, i.e., $v_n^{(2)} = -v_n^{(1)}$.

3. SBP Property and Energy Rate of a Single Element

By Assumption 1, the SBP property preserves the continuous divergence theorem exactly at the discrete level. Incorporating Assumption 2, the volume term inner product reduces to a boundary integral:

$$2\mathbf{q}^T \mathbf{WJ} \mathcal{L}_{\text{vol}}(\mathbf{q}) = \oint_{\partial T} v_n(q^-)^2 d\xi. \quad (3.27)$$

Multiplying the semi-discrete equation by $2\mathbf{q}^T \mathbf{W}$ and applying the above relation yields the energy rate for element T :

$$\begin{aligned} \frac{d}{dt} \|q\|_T^2 &= - \oint_{\partial T} v_n(q^-)^2 d\xi + \oint_{\partial T} 2q^-(f - f^*) d\xi \\ &= \oint_{\partial T} (-v_n(q^-)^2 + 2v_n(q^-)^2 - 2q^- f^*) d\xi \\ &= \oint_{\partial T} (v_n(q^-)^2 - 2q^- f^*) d\xi. \end{aligned} \quad (3.28)$$

4. Interface Energy Dissipation

The global energy rate sums over all elements. Consider an interior interface between adjacent elements Ω_1 and Ω_2 . Let $n^{(1)}$ be the outward normal of Ω_1 , with $v_n^{(1)} = n^{(1)} \cdot V$. By Assumption 3, $v_n^{(2)} = -v_n^{(1)}$. Let $f = f^*(q^{(1)}, q^{(2)})$ be the unique, anti-symmetric numerical flux. Summing their contributions gives:

$$\text{Rate}_{1+2} = \left[v_n^{(1)}(q^{(1)})^2 - 2q^{(1)}f^* \right] + \left[v_n^{(2)}(q^{(2)})^2 - 2q^{(2)}(-f^*) \right]. \quad (3.29)$$

Substituting $v_n^{(2)} = -v_n^{(1)}$ and factoring out f^* :

$$\text{Rate}_{1+2} = v_n^{(1)} \left((q^{(1)})^2 - (q^{(2)})^2 \right) - 2f^*(q^{(1)} - q^{(2)}). \quad (3.30)$$

For the upwind flux, substituting $2f^* = v_n^{(1)}(q^{(1)} + q^{(2)}) + |v_n^{(1)}|(q^{(1)} - q^{(2)})$ into the last term yields:

$$\begin{aligned} -2f^*(q^{(1)} - q^{(2)}) &= - \left[v_n^{(1)}(q^{(1)} + q^{(2)}) + |v_n^{(1)}|(q^{(1)} - q^{(2)}) \right] (q^{(1)} - q^{(2)}) \\ &= -v_n^{(1)} \left((q^{(1)})^2 - (q^{(2)})^2 \right) - |v_n^{(1)}|(q^{(1)} - q^{(2)})^2. \end{aligned} \quad (3.31)$$

Substituting this into Rate_{1+2} , the central difference part cancels exactly:

$$\text{Rate}_{1+2} = -|v_n^{(1)}| \left(q^{(1)} - q^{(2)} \right)^2 \leq 0. \quad (3.32)$$

Under these ideal assumptions, the equation confirms that the upwind flux provides strict energy dissipation at the interfaces, guaranteeing robust stability. However, in the actual Manifold DG implementation, these geometric identities are not exactly preserved.

5. Effect of Discrete Geometric and SBP Defects

In the actual manifold implementation, the metric Jacobian J and contravariant velocities $(\tilde{u}, \tilde{v})$ are non-polynomial rational functions. Consequently, their projection into the finite polynomial space introduces aliasing errors. The ideal conditions (Assumptions 1 to 3) are only approximately satisfied, leading to several defect terms. We define:

- **SBP Defect ($\mathcal{R}_{\text{SBP}}$):** Originating from the violation of the exact SBP identity (Assumption 1) on non-affine curved elements.
- **Geometric Defect ($\mathcal{R}_{\text{geo}}$):** Arising from the imperfect discrete metric identity and divergence-free condition (Assumption 2), meaning $\mathbf{D}_\xi(J\tilde{u}) + \mathbf{D}_\eta(J\tilde{v}) \neq \mathbf{0}$.
- **Projection Defect ($\mathcal{R}_{\text{proj}}$):** Caused by geometric aliasing, quadrature mismatches when evaluating nonlinear products, and imperfect anti-symmetry of normals (Assumption 3).

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是 aliasing
error.

Incorporating these defect terms, the realistic discrete energy evolution becomes:

$$\frac{dE_h}{dt} \leq -\mathcal{D}_{\text{interface}} + \mathcal{R}_{\text{SBP}} + \mathcal{R}_{\text{geo}} + \mathcal{R}_{\text{proj}}, \quad (3.33)$$

where $\mathcal{D}_{\text{interface}} = \sum |v_n^{(1)}| (q^{(1)} - q^{(2)})^2 \geq 0$ represents the stabilizing interface dissipation.

The exact discrete conservation is therefore compromised. If the geometric defects (which theoretically converge asymptotically as $\mathcal{O}(h^{k+1})$) exceed the stabilizing interface dissipation $\mathcal{D}_{\text{interface}}$, a drift in total mass and energy will be observed. This theoretical framework provides a direct explanation for the long-time mass conservation errors documented in the subsequent numerical experiments.

3.5 Time Integration

For time integration, we employ the fourth-order five-stage low-storage Runge–Kutta scheme proposed by Carpenter and Kennedy [1].

Given

$$du^{(0)} = 0, \quad Q^{(0)} = Q^n, \quad (3.34)$$

the stage updates are

$$du^{(k)} = A_k du^{(k-1)} + \Delta t \text{RHS}(Q^{(k-1)}), \quad (3.35)$$

$$Q^{(k)} = Q^{(k-1)} + B_k du^{(k)}. \quad (3.36)$$

The dimensionless time step is chosen as

$$\Delta t^* = \text{CFL} \cdot \frac{h_{\min}^*}{|\mathbf{u}^*|_{\max}(k+1)^2}. \quad (3.37)$$

The factor $(k+1)^2$ is adopted empirically based on standard stability behavior observed in high-order DG discretizations and should therefore be interpreted as a heuristic CFL scaling rather than a sharp theoretical stability bound.

Chapter 4

Results

This section verifies the proposed solver using two solid-body rotation configurations, simulating global advection phenomena in the Earth's atmosphere. The physical radius is set to $R = 6.37122 \times 10^6$ meters, and the advection is driven by a solid body rotation with a period of 12 days.

All numerical experiments in this study are performed using nodal polynomial spaces of order $k = 4$ for spatial approximation. Within the computational implementation, the dimensionless framework is applied:

- The dimensionless angular velocity is $\omega^* = \frac{2\pi}{12 \text{ days}} \times T_c$.
- The time step size Δt^* is determined by the dimensionless Courant–Friedrichs–Lewy (CFL) condition.
- The errors and integration times (in days) presented in the following figures are obtained by multiplying the dimensionless results back by the physical time scale T_c , facilitating a straightforward physical interpretation.

The velocity field $\mathbf{u}(\mathbf{x})$ is defined by the cross product of the angular velocity vector $\mathbf{\Omega}$ and the position vector $\mathbf{x} = (x, y, z)^T$:

$$\mathbf{u}(\mathbf{x}) = \mathbf{\Omega} \times \mathbf{x}. \quad (4.1)$$

Assuming the rotation axis is tilted by an angle α_0 relative to the polar axis, the angular velocity vector is defined as

$$\mathbf{\Omega} = \omega \begin{pmatrix} \sin \alpha_0 \\ 0 \\ \cos \alpha_0 \end{pmatrix}, \quad (4.2)$$

which yields the following Cartesian velocity components:

$$\mathbf{u}(x, y, z) = \begin{pmatrix} -\omega y \cos \alpha_0 \\ \omega x \cos \alpha_0 - \omega z \sin \alpha_0 \\ \omega y \sin \alpha_0 \end{pmatrix}. \quad (4.3)$$

Here, ω denotes the angular velocity magnitude. In the numerical experiments, two solid-body rotation configurations are considered with $\alpha_0 = 0$ and $\alpha_0 = \pi/4$.

For the initial condition, we employ a 3D Gaussian bell function mapped onto the spherical surface. Based on the dimensionless framework, the initial center of the bell is set at $\mathbf{x}_c = (0, 1, 0)^T$. To accurately represent the surface distribution, the initial scalar field $q_0(\mathbf{x})$ is prescribed using the geodesic distance r between $\mathbf{x}$ and $\mathbf{x}_c$:

$$q_0(\mathbf{x}) = \exp(-\sigma r^2), \quad (4.4)$$

where $r = \arccos(\mathbf{x} \cdot \mathbf{x}_c)$ is the great-circle distance on the unit sphere, and the width parameter is set to $\sigma = 10.0$.

Numerical convergence rates and mass conservation behavior are examined under various mesh resolutions defined by the subdivision parameter N . The spherical mesh is constructed by utilizing a regular octahedron as the base geometry. Each edge of the 8 base triangular faces is uniformly divided into N segments in the flat coordinate space. This subdivision yields N^2 smaller triangular elements per base face, resulting in a total of $K = 8N^2$ elements over the entire domain. The vertices of these sub-triangles are subsequently projected radially onto the spherical surface. The experiments are conducted using the following subdivision levels:

$$N = 4, 8, 16, 32. \quad (4.5)$$

To ensure geometric robustness and verify that the mesh does not degenerate during the non-linear mapping to the sphere, we compute the range of the surface Jacobian determinant (J) for each resolution. As shown in Table 4.1, the maximum and minimum values of J remain well within a reasonable bounding range and strictly positive throughout the refinement process. This confirms that the octahedral spherical mesh mapping maintains excellent quality without excessive distortion.

Table 4.1: Jacobian (J) range validation for polynomial degree $k = 4$

N (Divisions)	Min Jacobian (J)	Max Jacobian (J)
4	$2.51354471 \times 10^{-2}$	$8.11137610 \times 10^{-2}$
8	$5.00646667 \times 10^{-3}$	$2.02962046 \times 10^{-2}$
16	$1.10652039 \times 10^{-3}$	$5.07434750 \times 10^{-3}$
32	$2.59893344 \times 10^{-4}$	$1.26859158 \times 10^{-3}$

To quantitatively evaluate the numerical accuracy, the absolute L^2 and L^∞ error norms are computed over the dimensionless unit spherical domain $\Omega = \mathbb{S}^*$ as follows:

$$L_{\text{error}}^2 = \sqrt{\int_{\Omega} (q_{\text{num}} - q_{\text{exact}})^2 d\Omega}, \quad (4.6)$$

$$L_{\text{error}}^\infty = \max_{\mathbf{x} \in \Omega} |q_{\text{num}}(\mathbf{x}) - q_{\text{exact}}(\mathbf{x})|, \quad (4.7)$$

where q_{exact} is the exact analytical solution given by the advected initial condition. As established in Section 2.1, both the grid parameter h and the error norms reported are completely decoupled from the physical scaling R .

All experiments are conducted with $\text{CFL} = 0.5$. Both cases are integrated to a final physical time of $t = 12$ days.

4.1 Case 1 (Solid body wind $\alpha_0 = 0$)

This case corresponds to zonal solid-body rotation aligned with the equator.

Table 4.2: Convergence results for Case 1

N	h	L^2 error	L^∞ error	L^2 order	L^∞ order
4	2.5000×10^{-1}	5.802×10^{-3}	3.076×10^{-2}	—	—
8	1.2500×10^{-1}	3.241×10^{-4}	3.124×10^{-3}	4.2	3.3
16	6.2500×10^{-2}	1.573×10^{-5}	1.788×10^{-4}	4.4	4.1
32	3.1250×10^{-2}	8.179×10^{-7}	9.958×10^{-6}	4.3	4.2

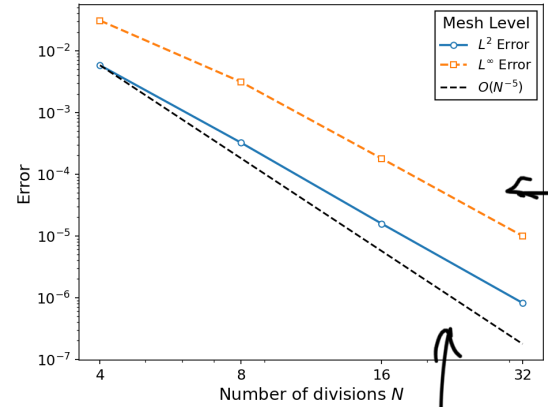
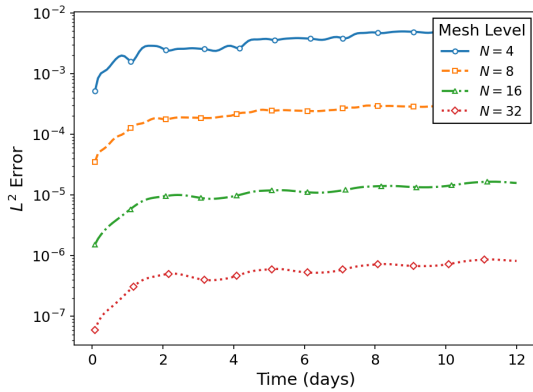
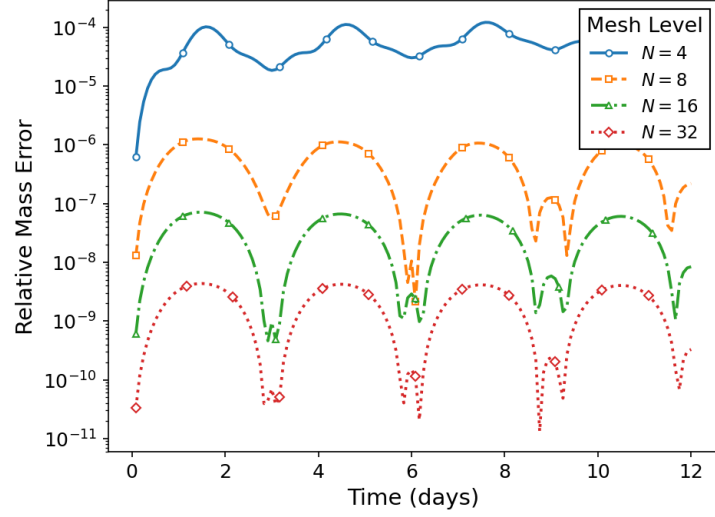


Figure 4.1: Error and convergence plots for Case 1.

證明結果？

也把 $O(N^5)$ 畫一下

To quantitatively analyze the numerical pathology regarding the lack of exact discrete geometric conservation (DGCL) discussed previously, Figure 4.2 illustrates the temporal evolution of the relative mass error ($\Delta M/M_0$) over the integration period up to $t = 12$ days. The drift in mass conservation serves as empirical evidence of the geometric aliasing effect introduced by the metric approximation.



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收敛的级

Figure 4.2: Relative mass error evolution over time for Case 1, highlighting the temporal drift in discrete conservation.

4.2 Case 2 (Solid body wind $\alpha_0 = \pi/4$)

This test involves inclined solid-body rotation crossing multiple Octahedral element interfaces and vertices.

Table 4.3: Convergence results for Case 2

N	h	L^2 error	L^∞ error	L^2 order	L^∞ order
4	2.5000×10^{-1}	7.901×10^{-3}	3.975×10^{-2}	—	—
8	1.2500×10^{-1}	1.790×10^{-4}	1.359×10^{-3}	5.5	4.9
16	6.2500×10^{-2}	5.988×10^{-6}	3.784×10^{-5}	4.9	5.2
32	3.1250×10^{-2}	3.862×10^{-7}	2.061×10^{-6}	4.0	4.2

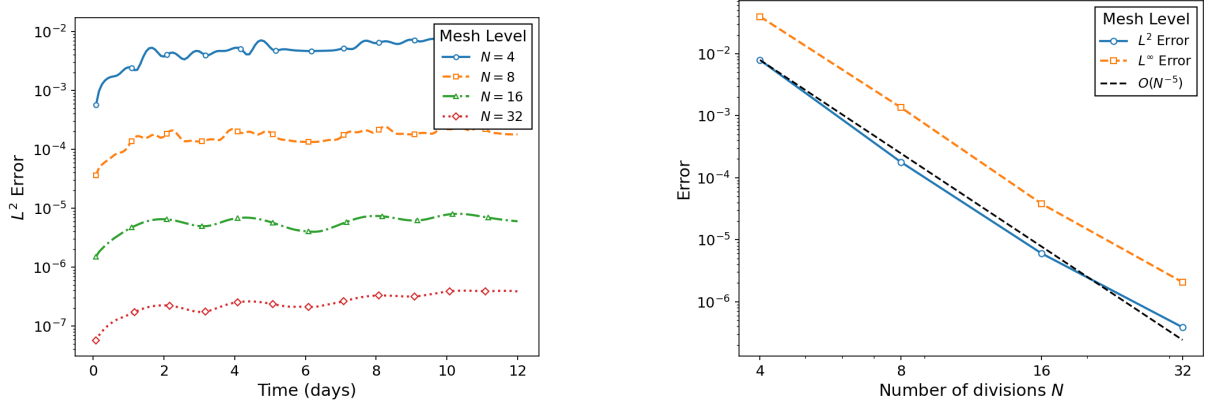


Figure 4.3: Error and convergence plots for Case 2.

Similar to Case 1, the temporal evolution of the relative mass error for the inclined rotation configuration is presented in Figure 4.4.

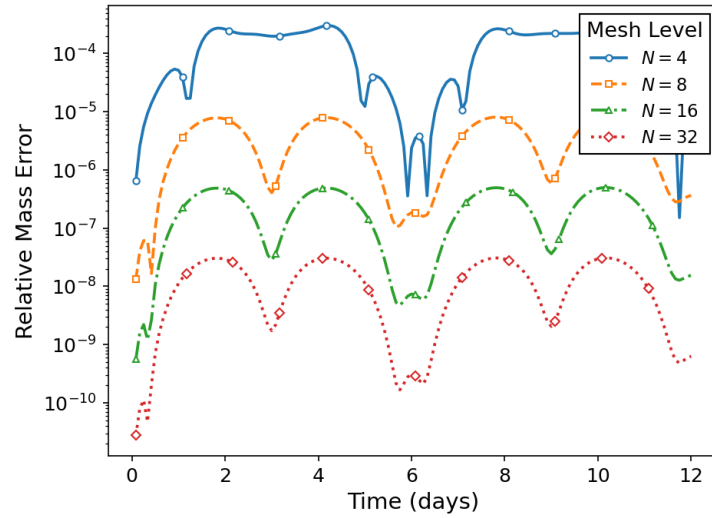


Figure 4.4: Relative mass error evolution over time for Case 2.

结果说明

Chapter 5

Conclusion

This study develops a three-dimensional Manifold Discontinuous Galerkin formulation for scalar advection problems on spherical geometries. By employing a directly embedded Octahedral manifold mesh in $\mathbb{R}^3$, the proposed method avoids the pole singularities and topological patch connections commonly encountered in traditional spherical parameterizations.

Numerical experiments involving solid-body rotation demonstrate stable fourth- to fifth-order convergence behavior in both L_2 and L_∞ norms. These results confirm that the proposed manifold-based DG formulation provides robust and accurate transport simulations on curved spherical geometries.

The study additionally suggests a geometric limitation of standard nodal DG formulations on curved manifold meshes: exact discrete conservation is generally not preserved. Although the continuous geometric conservation law (GCL) is satisfied analytically by the underlying manifold geometry, the discrete geometric conservation law (DGCL) is violated due to geometric truncation and projection errors.

More specifically, the following mechanisms contribute to structured error decomposition:

1. The metric Jacobian and contravariant velocity components are non-polynomial functions. Their projection into degree- k nodal polynomial spaces introduces geometric aliasing errors, causing the SBP structure to hold only approximately rather than exactly.
2. The discrete metric identities are only approximately satisfied because the projected metric terms cannot exactly preserve continuous differential geometric relations on curved elements.
3. Independent geometric mappings of adjacent curved elements generate slight pointwise mismatches in interface normals, leading to imperfect anti-symmetry of the discrete normal velocity.
4. The discrete incompressibility condition is not preserved exactly because the contravariant metric terms and projected velocity fields are only approximately represented in the finite-dimensional polynomial space.

Consequently, the present formulation should be interpreted as a curved-manifold DG discretization that is compatible with the affine-element SBP structure up to geometric approximation errors, rather than an exactly conservative curved-element SBP discretization.

~~5.1 Discussion~~ この部分 section

The numerical evidence suggests that the dominant source of conservation error originates from the interaction between curved geometry and polynomial truncation. In particular, the discrete operators are algebraically exact only within the finite polynomial approximation space, whereas the manifold metric quantities generated by spherical normalization are rational functions.

Therefore, even under over-integration, the nonlinear products

$$J\tilde{u}q, \quad J\tilde{v}q \tag{5.1}$$

are not exactly representable in the finite polynomial space. This leads to persistent geometric aliasing and approximate discrete metric identities.

These observations indicate that future conservative manifold DG methods must simultaneously address:

1. exact or compatible geometric representations,
2. SBP-compatible metric discretizations,
3. discrete metric identity preservation,
4. interface-normal consistency across curved elements.

~~5.2 Future Work~~

Future work will focus on constructing a strictly conservative and entropy-stable manifold DG framework.

First, we plan to investigate exact or analytically consistent geometric evaluations for the Jacobian and metric terms in order to reduce projection-induced aliasing errors.

Second, we aim to develop modified differentiation operators satisfying curved-element SBP identities in a weak or projected sense.

Third, we plan to derive quantitative DGCL error estimates consistent with the underlying polynomial approximation accuracy while accounting for metric projections, quadrature effects, and geometric aliasing contributions.

Finally, future developments will explore entropy-stable manifold formulations based on compatible geometric projections, intrinsic surface calculus, and conservative metric constructions suitable for large-scale atmospheric and oceanic simulations on spherical manifolds.

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